

Ms. Hana Kouiriti

Moroccan Citizen | hkouiriti@gmail.com | (+86) 177 9160 8707 | Address: Chang'an, 710100, Xi'an, China
Portfolio website: <https://hanaportfolio.netlify.app/> | LinkedIn: <https://www.linkedin.com/in/hana-kouiriti/> | GitHub: github.com/bla999ckt

PROFESSIONAL SUMMARY

Results-driven Computer Scientist with 5+ years of programming experience and expertise in Artificial Intelligence/Machine Learning (AI/ML), embedded systems, and full-stack development. Currently an undergraduate student at Northwestern Polytechnical University (NPU), awarded the First-Type Presidential Scholarship for academic excellence. Skilled in C, C++, Python, Java, and Verilog, with hands-on experience in Field-Programmable Gate Array (FPGA) development, autonomous systems, and AI model training. Actively engaged in open-source contributions and AI research, aiming to bridge the gap between theory and real-world applications.

EDUCATION

Northwestern Polytechnical University

Xi'an, China

B.Sc. in Computer Science

Expected Graduation: July 2027

Planned: M.Sc. in Artificial Intelligence

- **GPA:** 3.745/4.0
- **Relevant Coursework:** Data Structures & Algorithms, AI: Fundamentals and Practical Applications, Probability Theory
- **Languages:** Arabic (Native), English (Fluent), French (Fluent), Chinese (HSK3)

EXPERIENCE

AI Model Development – Heart Disease Prediction

Personal Project | Remote | 2024

- Trained a neural network model on a dataset of 10,000+ patient records using Python and TensorFlow.
- Achieved 92% prediction accuracy (outperforming baseline models by 15%) via feature engineering and hyperparameter tuning.
- Reduced false positives by 20% through stratified sampling and Synthetic Minority Over-sampling Technique (SMOTE) for class imbalance mitigation.

AI Assistant for University Students (PearlAi)

University Project | China | 2024

- Fine-tuned Qwen3.5 32b on 1500+ academic Frequently Asked Questions (FAQs), reducing response latency to <1 seconds via model quantization.
- Deployed chatbot on Alibaba Cloud server with 1,200+ monthly active users; improved student query resolution rate by 35%.
- Built a React-based frontend with a 95% Lighthouse accessibility score and <2s load time.

Autonomous Airship Prototype

Team Research | China | 2024–Present

- Programmed flight stabilization algorithms in C++ for Matek F405, achieving 95% stability in wind simulations.
- Reduced navigation path deviation by 40% using Proportional-Integral-Derivative (PID) controllers and Kalman filters.
- Integrated FPGA-based control logic, cutting processing latency by 30% compared to baseline Microcontroller Unit (MCU).

Personal Portfolio Website

Personal Project | 2024

- Built with React and Tailwind CSS, achieving a 100/100 Google PageSpeed score via lazy loading and asset optimization.

- Increased user engagement by 50% with interactive animations and a project gallery (15+ projects showcased).

FPGA-Based Digital Logic Projects

Coursework & Independent Projects | 2024

- Designed an 8-bit Arithmetic Logic Unit (ALU) with 16 operations (arithmetic/logic), achieving 200 MHz clock speed on Xilinx FPGA.
- Reduced adder propagation delay by 25% using carry-lookahead techniques in Verilog.
- Automated testbench verification with 98% coverage across 50+ test cases in ModelSim.

Competitive Programming & Open-Source

LeetCode, GitHub | 2023–Present

- Solved 400+ problems, specializing in dynamic programming and graph algorithms.
- Contributed to 6+ open-source projects, including a VS Code extension(BlackT codestyle) with 5,000+ downloads.

Game Development – Breakout & ShellStorm

Personal Projects | 2023

- Developed Breakout with 30+ levels and dynamic difficulty, achieving 60 Frames Per Second (FPS) on low-end hardware using OpenGL.
- Designed ShellStorm with 10+ enemy wave types; optimized collision detection to handle 1,000+ entities with <5ms latency.

Interactive Visualization Projects

Personal Projects | 2023–Present

- Created a Python-based whale animation with 60 FPS smooth cursor tracking using Pygame’s vector math.
- Built a CSS lightbulb effect with <10ms render time and mobile-responsive design (4.5/5 user rating).

ACTIVITIES & LEADERSHIP

BlackPearl | Co-founder & AI Lead | 2023–Present

- Led a team of 5 developers to build a university AI assistant, adopted by The International School of NPU (about 5000 users).

High School Programming Club | Leader | 2021– 2023

- Mentored 20+ students, increased competition participation by 60% (team won 2 regional hackathons).

Recycling & Sustainability Fair | Participant | 2023

- Awarded a Certificate of Appreciation for creating a recycling project with a database and a Flask app to track and visualize waste reduction.

AWARDS & CERTIFICATES

NPU Presidential Scholarship – First Prize

Awarded to the Top 10% of Academic Performers

- Recognized for academic excellence among the top 10% of students in my cohort at Northwestern Polytechnical University.

2nd Prize – AI Application Innovation and Practice Competition

Team Leader, PearlAI Project (BlackPearl Team)

- Achieved second place in an AI innovation competition for developing PearlAI, a university assistant designed to support international students using large language models.

VISION Global Competence Association – Activity Participation Certificate

Northwestern Polytechnical University

- Acknowledged for actively assisting in the organization and execution of intercultural events and student activities that promoted global competence and engagement.

Academic Excellence Certificates – Middle & High School

Alal El Fassi – Al Irfan High School

- Recognized for consistently ranking among the top students and graduating first in my major.

SKILLS

AI/ML: Deep Learning (TensorFlow, PyTorch, Keras), Machine Learning (Scikit-learn, Pandas, NumPy), Natural Language Processing (transformers, embeddings, Large Language Model (LLM) fine-tuning), Computer Vision (OpenCV, image recognition, object detection), Model Optimization (quantization, pruning, deployment on edge/FPGA).

Programming: Python (primary), C/C++, Java, MATLAB, SQL.

Software & Frameworks: Flask, FastAPI, Docker, Git, Bootstrap, Tailwind, Pygame.

Web Technologies: JavaScript, HTML, CSS.